Why Laser Beams Are the Hottest New Tech in Defense

Podcast: Odd lots

AI Executive Summary

1. **Core Thesis** Directed Energy Weapons (DEWs), encompassing high-energy lasers and high-powered microwaves, are transitioning from theoretical research into practical defensive applications to address asymmetric threats like drone swarms and hypersonic missiles. While these systems offer a sustainable "deep magazine" advantage over kinetic munitions by reducing cost-per-shot and replenishment needs, they remain constrained by significant hurdles in power generation (size, weight, and power trade-offs), critical mineral supply chains, and environmental dependencies. The defense industrial complex is actively shifting toward an integrated air and missile defense architecture that combines elite interceptors with emerging directed energy capabilities to maintain economic sustainability and operational superiority against adversaries who possess high volumes of low-cost munitions.

2. **Key Arguments & Data**

3. **Weaponization Differences:** High-powered microwaves act as an "electronic shotgun," capable of frying electronics in swarms or hitting multiple targets simultaneously, whereas lasers function as "sniper rifles" that concentrate heat to physically destroy single targets (e.g., tanks) through material burn-through.
4. **Power and Deployment Constraints:** Current 50kW laser platforms rely on generator power; future high-end systems aim for 1MW by 2030. A major bottleneck is creating deployable units with sufficient mass vs. magazine capacity, as mobile generators are required to replace shore power dependencies for kinetic warfare readiness.
5. **Economic Shift:** The cost-per-shot for directed energy is significantly lower than expensive interceptors (e.g., Lockheed Martin PAC-3 or RTX SM-6) costing millions, but deployment requires more upfront investment in infrastructure and critical minerals compared to traditional munitions.
6. **Market Forecast:** The Directed Energy market is projected to grow from $6-8 billion currently over the next decade, supported by over

$15 billion already spent on R&D. Demand signals indicate a need for "middle ground" solutions (e.g., Coaspire, Lightforce) filling the gap between cheap interceptors and exquisite missiles.

7. **Chemical vs. Electrochemical Energy:** Early development involved chemical lasers (e.g., Oxygen-Iodine system used on the YAL-1 Boeing 747), which generate energy via chemical reaction rather than electricity but remain large-scale and logistically intensive, leading to a shift toward diode-based electrochemical systems.
8. **Critical Mineral Supply Chain:** Production relies heavily on rare earth elements such as Terbium and Dysprosium, with China holding 85-90% of the supply chain. Dual-use applications complicate industrial decisions as these minerals are also vital for civilian drones and commercial technologies.
9. **Layered Defense Strategy:** Effective defense requires a mix of capabilities: Exquisite munitions (hypersonic interceptors like Arrow/THAAD) for high-level threats, paired with cheaper systems (Iron Dome/Iron Beam, Vampire/APKWS, Coyote) for mass targets like the Shahid-136 drone.
10. **AI and Human-in-the-Loop:** AI currently assists in imagery tagging, normalization, and noise reduction but remains strictly under "human-in-the-loop" for target execution authority due to trust models and safety concerns regarding autonomous destruction of friendly force.

11. **Operational Depletion vs. Innovation:** The U.S. military is facing munitions depletion (e.g., Patriot interceptors) due to sustained conflict with asymmetric threats, driving a push for multi-year framework agreements with defense primes to increase stockpiles and capacity while startups handle scalable, lower-cost production lines.

12. **Actionable Takeaways**

13. **Track Supply Chain Vulnerabilities:** Investors and industry analysts should monitor supply chain independence strategies regarding critical minerals (Terbium, Dysprosium), specifically noting partnerships like MP Materials for onshore manufacturing capabilities within the U.S. defense industrial base.

14. **Evaluate Defense Sector Dynamics:** Assess investment opportunities in the "middle ground" defense market, focusing on emerging startups (e.g., Andurion, Lightforce) that offer scalable, cost-effective alternatives to traditional prime contractors for drone defense, while monitoring prime contract growth for high-end interceptors.
15. **Monitor R&D Milestones:** Keep an eye on the Department of Defense's timeline to achieve 1MW laser capabilities by 2030, as this marks the threshold for effective hypersonic counter-measures and indicates a shift from experimental procurement to operational deployment.
16. **Analyze Threat Signatures:** Understand that layered defense systems must dynamically match interceptor signatures (boost, glide, terminal phase) to threat types; using an expensive interceptor against a low-value drone is economically inefficient and highlights the strategic value of electronic warfare and laser options.
17. **Assess AI Reliability:** In defense contexts, prioritize technology where human judgment remains central for kill decisions to avoid escalation risks, while leveraging AI primarily for data processing and threat detection rather than autonomous engagement.

Full Transcript

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have some really great conversations planned. So go ahead and get your tickets. You can find those over at bloomberg.com forward slash odd lots or click the link below in the show notes and come and say hi when you're there. You know, normally you and I talk for a couple minutes and then I introduce the guest. Yes. I'm just going to introduce the guest right away and say today we are speaking with Colonel Sanders. I actually I have some sympathy for. this issue of people with famous military names because my dad was in the Air Force and his first name is Tom. Can you guess what the highest rank was that he actually achieved in the Air Force? Oh, major. Yep. And he always claimed he was under promoted because people found it funny to have Major Tom in the Air Force. Of course, you have to ask the question of whether he was over promoted to get to Major Tom. I bet I bet he never saw that story, but it's just as plausible.

That's right. OK, so we'll have to ask our guest whether he was overpromoted to Colonel or underpromoted to Brigadier General. But we're going to be speaking with Wayne Sanders. He's the Senior Defense Research Analyst over at Bloomberg Intelligence. And we're going to be

talking about laser beams and microwave weapons. Great. I don't know anything about laser beams or microwave weapons. This is all going to be new to me. I have seen in sci-fi movies that, you know, In theory, you could shoot a laser. Pew, pew, pew. But like, I don't know if that's a real thing, or if that's just like a sort of cinematized version of what a visual weapon could be like. I have no idea. Evidently it is real-ish because there is research on laser weapons and energy weapons and stuff, but I don't know anything about how this works. So two things I learned in the course of preparing for this conversation. So number one.

Both these things fall under the umbrella of directed energy weapons, which sounds directed energy sounds like something my yoga instructor would say if I had a yoga instructor, which I don't. But at the same time, people have been talking about these forever. They've been in development for literally decades now. And I think the big question is whether or not we're hitting a moment when they might actually become reality. So we know, for instance, that Israel is getting ready to deploy its iron beam. project. I think there's some discussion of whether it should be called laser dome instead of iron beam, which is kind of funny. But at the same time, we know that the US is ballistic missile supply has been dwindling because you know, it doesn't really make a lot of sense to use like weapons that cost millions of dollars to shoot down thousands of drones that are made out of styrofoam and plywood in some cases. Yeah, it's clearly a lot of anxiety about just the sheer number of munitions in any context etc we've been obviously the world is exhausting a lot you read a lot of stories if we could replace a bunch of it with lasers maybe that could be a solution it's interesting to think you know i sort of put this in a category of certainly core science technologies that have been the next big thing for a long time you probably remember what was it called like nanotech remember when like nanotech was gonna be I don't know what's going on with nanotech robots. Another one is quantum computing. There does seem to be a lot of progress, but also when I talk to someone in quantum computer, it still seems like we got a ways to go, etc. And so the way you describe this sort of high energy weaponry, I'm curious whether it's one of these things that definitely works in theory and ambiguously works in practice. Yeah, so we're going to try to disentangle all of those themes. There's also a really interesting supply chain attached to some of these weapons, obviously. So I'm happy to say we

do in fact have the perfect guest. We're going to be speaking with Wayne Sanders. As I said, he's the senior defense research analyst at Bloomberg Intelligence and formerly a colonel in the US Army. So Wayne, thank you so much for coming on all thoughts. Thanks so much for having me. It's great to be here. Why don't you give us the I guess quick overview of your military career because you come to us not just with you know analytical bonafides but with some actual operational expertise. No thank you again so I graduated from West Point in 2002. I went military intelligence so I had a background in tactical intelligence early on and then obviously the war kicked off.

I transitioned over to signals intelligence where I spend a lot of time looking at computer network exploitation was the next best thing that was out there, right? So signals intelligence where you're looking at cell phones and other pieces quickly transitioned into the computer side of the house. I then ended up at the national security agency for quite a while with US cyber command. And I transitioned over to becoming an offensive cyberspace operations officer, what's known as a 17 alpha, where I got to spend the rest of my career working on very hard problems for the US military to gain and maintain access to adversary networks. And then when called upon deliver those capabilities. My last job in the army was I was the chief of staff for the undersecretary defense for research and engineering honorable Heidi shoe, where I got to spend a lot of time.

on critical technology areas specifically being developed out. So that's really where I got a chance to geek out and look at what some of the next research and development areas are in these weapon systems. And that kind of translated right over into coming over to Bloomberg Intelligence, where I work on the defense industrial base, emerging technologies, weapon systems, and who are the companies and the trends that are out there for global funding as well. That was amazing. That was perfect and I'm extremely riveted and excited about this conversation. Talk to us a little bit about your remit here and like the main sort of gist of the work that you do when bringing intelligence. Yeah, absolutely. One of the areas that we were really that Bloomberg was really looking forward to to moving forward with was going a step further in the aerospace and defense industry area, not just the primes, not just looking specifically at the companies, but

looking at the ecosystem as a whole. So not just OK, we're looking at Lockheed Martin and the PAC three interceptors and where some of those margins are but also then looking at the air defense market as a whole looking at as you mentioned earlier in terms of laser dome versus golden dome or you know.

when you start looking at those pieces it's okay there's something called integrated air and missile defense there's an entire program there is a big difference between an exquisite munition interceptor like a pack three and there's much difference between that and a high-energy laser that's going to shoot a potential drone or a ballistic missile out of the sky So when you put all those pieces together, what does the air defense market look like? What does hypersonics look like? What does semiconductors as part of a supply chain concern for the defense industrial base? Shipbuilding, you name it. These are some of the areas that we cover down on and be able to focus on everything from budgets, components, supply chain, regional components. Really, you look at the U.S. allies, if you will.

adversary considerations that are driving where those those markets may be going and then what's next in the queue so when you look at air defense a lot of the stuff up until now a lot of the direct energy stuff we're going to talk about today is some of those futures piece where is it in the r&d line for research and development and then where is it going and when are we transitioning into procurement All right, so before we talk about the role of directed energy in I guess the military industrial complex and defensive strategy more broadly, let's start really, really simply. And let's just can you can you define what this actually is when we talk about directed energy weapons like laser beams or microwave systems?

What exactly are we talking about here? Absolutely. Absolutely. You know, a lot of people call it electronic warfare. That's a name that a lot of times ends up being synonymous as well, because you look at electronic attack platforms. You are trying to take specific harnessed energy and you are trying to use it in an offensive way to cause some type of degradation or destruction to something else. And so when you look at a laser, a laser is essentially a precision weapon. It concentrates energy into a narrow. beam and typically attacks one target at a time. But when you look at high

powered microwaves, those weapons attack electronics. It's a larger beam that can potentially affect multiple systems simultaneously. So the government accountability office notes that high powered microwave weapons can generate peak powers up to like 100 megawatts. But comparing that number directly to continuous wave laser kilowatt ratings is misleading because of the characteristics that are fundamentally different.

What I mean by that is a laser is a sniper rifle. And a microwave is the electronic shotgun. Think about shooting something with bird shots because a laser, you're trying to scale that power and the microwave is trying to scale the effect. So that's why you're looking at drone swarms. And what can you potentially do for that? Right. So the microwave, fire the weapon or whatever it may well get. And then suddenly all the drones don't work. where because their electronics have been fried or something like that. And then a laser is something that could theoretically be pointed at, I don't know, a tank or something like that, and then disable it. Like, is it sort of if let's just say we're talking about the lasers.

Is it what I have in mind only from having watching a movie where it's like, there is a beam and then it's bad to be on the other side of the beam? That's pretty accurate. You know, when you start looking at how strong those lasers are, you are trying to direct that energy into a single point that is going to burn a hole. It's trying to degrade, right? And not so much on the tank. And it's heat. Sorry. Yeah. just to be clear it's that heat that's the destructive force there in the lake. Correct it's it's the heat and it's the energy that's being pushed through right so if you get that hot enough you actually can burn through that material specifically for that when you look at it normally you're looking at the difference between a drone swarm and potentially like a ballistic missile because a drone swarm perspective you're going to want a high powered microwave you want something that is a wider a wider range you want something that's going to be able to hit multiple targets at the same time and cause them to degrade and it may not they're not going to blow up but they might fall out of the sky something's not going to work on those but if you're pointing a laser directly at a single ballistic missile that's coming at you you want to be able to destroy that thing you want to be able to knock that out of the sky at that point in time.

Where does the energy for these things actually come from? Because when we talk about laser beams by definition, that's a densely concentrated beam of energy. And I can't imagine that militaries are plugging these things into the grid, although maybe they are, or maybe they're rolling around with like giant battery packs or something like that. But where does the energy actually come from? So the answer is yes to both of your questions actually and it kind of goes into some of the history of how lasers came to be and to work those out is you normally call that shore power. A lot of the technology behind lasers and all that requires a huge amount of power that's going to create that specific spot beam that you're actually trying to create that directed beam of energy you know starts in these diodes but you have to be able to then power those there's an awful lot of power in the past that was required in order to create that level of beam that you're trying to be able to push out you want to keep it refined small enough so that it is directed in the area that you want so it's not going out and refracting off and hitting other targets right but you you start looking at that is it's one of the reasons why back in the day some of these lasers were the size of buildings because you required that much energy to be able to create that beam, to be able to create that pulse. So you're seeing actually the difference really in size, weight, and power. You'll hear that quite a bit when you're dealing in this type of environment is the swap characteristics, right? Size, weight, and power. Because obviously if you're trying to create something that's deployable, you can't just plug it into a grid and say, okay, all right, well, because now you're static.

Yeah, and so it's great if you're using something for a base defense Okay, and you have you have clean enough power and everything else to be able to plug that in to be able to use But the deployability aspect of where a lot of the militaries are going right now is I need to be able to throw something onto a truck I need to be able to throw something onto a ship and I need to be able to plug that thing into some type of shore power so that I have the ability to have regenerative electronic capabilities to create those beams and increase what we normally call is the difference between mass versus magazine because at that point in time you want to have additional shots and you want to bring down the cost per shot and this is one of those areas where you can do that.

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So is the technological advancement that's brought us to the current time where we can talk about laser beams as a potential reality on the field, is that all due to energy advancement? What has gotten us to this point? Yeah, no, absolutely. I think so. First off, it really started with the physics problem. If you go back even to the 60s, right? When the laser arrived, the military immediately recognized that it had potential and the early work seemingly, you know, is a revolutionary idea that you could put energy onto a target and essentially do it at the speed of light. Then you had the Star Wars era where I identify specifically where they're trying to produce.

extremely powerful chemical lasers. And this is where you start looking into the difference between the chemical laser problem back then required a lot of energy. It also required a very large facility because of how big these things were. So if you go back to like the 1990s and the 2000s, US Israel had something called the tactical high energy laser, the fell. It demonstrated that a laser could actually destroy rockets and artillery rounds in flight. There was also something from DARPA called Miracle. the mid-infrared advanced chemical laser. I had to write it down because I didn't remember what it actually said. But it destroyed live short-range rockets as well. But the chemical laser architecture remained really large and logistically intensive, so the program ended in 2006. I looked this up as soon as you said chemical laser because I went, what the heck is that? It says it gets its light energy directly from a chemical reaction instead of using electricity. Yes. Is that right? Yep.

What kind of chemical reaction? Now I'm like imagining soldiers with a big like that of chemicals and like, yeah, I'm mixing them together. And then suddenly it goes boom and shoots into the sky. No, not so much like that.

But it normally is something in a containerized area. There's real science. If you remember that movie with Val Kilmer back in the day, being able to kind of create there's some some oxygen iodine laser. you know type reaction where it's something that you can actually control it's something that you can put inside of a a mechanism and you can control it from there yeah so you don't have to actually it's not like us in chemistry class you know putting stuff into a beaker and then pouring it in so it is something that's the oxygen iodine laser has been used in the past. Also, they put this thing onto a Boeing 747 at one point in time called the YAL-1. But the problem there is, once again, you're dealing with now you have a chemical reaction that's taking place on board of a ship. And oh, they were also, by the way, trying to do this to fly near where you think a ballistic missile launch is going to go. So they can now chase the missile and shoot this thing out of the sky. So you can fly it down from a plane. Absolutely. And well, it's risky.

Yeah, I'm gonna ask like a sort of dumb question, but that may be it could lead us to an interesting question. Like, do we have a gun in existence that looks like something where you hold it like it resembles outside of normal gun and fires a laser beam in a military capacity? Not not in the way that you're you're thinking. I mean, we have we have some areas like you know, there's the Helios platform that's on the USS Preble. So if you want to think of a very large ship gun that has the capability of doing that, you know, the Navy does have that. If you're looking for a handheld rifle, you know, something like a laser or phaser kind of thing that you've seen in science fiction and all that, not so much.

where you would actually see the directed beam. We do have drone guns that can be used for RF and be able to push out to knock drones out of the sky though. Well, so I figured that that laser gun that I'm imagining in my head doesn't exist, but what would be the constraints to its existing? So, okay, we can, or at least currently as of 2026, what would be the main bottleneck or thing we would need to solve? I imagine the thing we would need to solve would be building an insanely dense battery or something like that. But what would be the thing we would need to solve to have the laser weapons that we see on screen? Yeah, I would say it's power generation and compression when you start looking at what you have to be able to do. The

reason why it's still existing really striker vehicles right now have there are a couple of platforms that are out there.

specifically for this with a high-energy laser on there, but you're talking about 50 kilowatts of power. That's something that's being generated based off of the generator power that's already put on a Stryker vehicle. That's part of that shore power that's in there. Like you're talking about, whether it's batteries or whether or not it is some type of power generation mechanism is going to be required in order to create the level of beam that's necessary. If you only have a five kilowatt laser beam at that point in time, but it requires 15 or 25 kilowatts of energy in that beam to be able to knock a drone down, then at that point in time, it's not going to be effective. So you have to make sure that you are pairing the type of capabilities, the power generation that you can push out versus the threat that you're actually trying to mitigate. You know, Tracy, whenever we talk defense tech in any capacity, there's always this weird thing where it's like.

We're talking about machines that kill people, right? And it's what I'm reading. They I'm on the Lockheed Martin looking at their Helios integrated weapon system and it's going through the specs and it's like deep magazine, low cost per kill, speed of light delivery. It's like it's gear, but you know, these are the these are the metrics. Yes. Yeah, it's creepy for sure. So actually one thing I wanted to ask Joe just touched on this deep magazine. idea. And sometimes when you read about the potential for directed energy weapons, people talk about like, Oh, there are more cost effective tactic against things like cheap drones. So instead of firing off a pack three that costs millions of dollars, you fire off a laser beam that is, you know, renewable, you can keep firing or you send off microwave beams and keep doing that. But Is comparing directed energy weapons to something like ballistic missiles, is that the right comparison? Because I also hear about things like coyote or there's something called vampire, which I would love to learn more about or Skynex, like systems that can do a similar thing but are also cheaper than ballistic missiles.

Absolutely. So one of the reasons that you look for high-powered microwaves, high-energy laser solutions, this directed energy platform is because the cost per shot is very low. As long as you have the stream of

energy necessary to be able to use it, it's great. The problem is that you're waiting for something that is trying to kill you. to get close enough to the target so that you can hit this thing with high-powered microwaves or a high-energy laser to be able to knock that out. And oh, by the way, when you have adversary capabilities that are traveling at Mach 5, you're traveling at 4,000 miles an hour, you probably don't want to hope that you can put this laser beam exactly where you want it to be in order to do that. So what you're really looking for is there's layered air defense. I referred to it earlier as integrated air and missile defense.

And this is where it's really important because it's super relevant to what we're talking about right now, especially when you look at investment as well. You have to remain, you have to have exquisite missiles. You have to have your Lockheed PAC-3 interceptors. You have to have your RTX SM-3s and SM-6s. You cannot get rid of those because you have adversaries who have hypersonic weapons. If a hypersonic weapon is launched, you're going to want to use one of your more exquisite capabilities to make sure that you can knock that thing out of the sky and it doesn't kill your soldiers, your airmen, sailors, and Marines and your guardians. You want to make sure that that doesn't take place. So the exquisite missiles are part of this conversation. But the biggest piece, like you're saying right now, is you're identifying that you can win every engagement. You can win every engagement.

defensively and still lose economically. And that's what we're seeing a lot in Iran right now is they're just starting to just run us out of some of these exquisite interceptors and munitions. And so the, as you mentioned with coyote and you mentioned with vampire and BA has a system called APKWS, which is the advanced precision kill weapon system. Ryan Mital has Skynex and Skyranger. These are cheaper interceptors in guns. So think of these as almost a mid-range solution for those to say, okay, I don't want to use a 4 million because you can see the radar systems themselves, if something is launched and coming at you, it will put together a signature. It will say, this is this type of system. It's being shot at us at this type of speed. It's going to have a boost phase, glide phase, terminal phase on some of these capabilities, right?

depending on what those things are, it's important to be able to say, okay, well, based off of the threat, I want to pair it with a specific type of capability. That makes sense. And so if you're being shot at, if a drone swarm is coming at you and it's the Shahid-136 is out of Iran, You don't need to shoot it down with the Pac-3 interceptor. You may be looking for that Vampire APKBWS partnership. You may want to use some of these Coyote Block IIs. You want to go and look at some of these systems and say, that's what I want to use. If it's going slow enough or you have enough of them, you may want to then partner that with a high-powered microwave to knock them down. You don't want to give up any ground at all in terms of creation of these different capabilities. These end up being additive.

And Israel does it better than anyone. When you look at Israel, if you have an intercontinental ballistic missile threat that's coming your way, they're going to use their Arrow 2, Arrow 3 systems, which are exo-atmospheric. They are going to try and shoot this thing down as far away from you as possible based off of how serious the threat is. If it's something else that's a little bit shorter than that, they have a system called David's Sling. And David's Sling goes farther than Iron Dome. Iron Dome is that next layer that you use and you hear about it all the time between Iran and Israel and the 12 they were as well as during Epic Fury and into the systems now is they continue to use Iron Dome and now they're adding in Iron Beam as well. So what it's done is it gives the commander options so that you don't always have to use your exquisite munitions. You don't have to bankrupt yourself to do it.

If you have this integrated architecture that goes across all these different weapon systems for the reasons you just laid out, who actually makes or owns I guess the kill decision? Like who decides what deployment of all those different things actually looks like against which targets? And I imagine most of the time you're operating under like highly pressurized time environments, right? Yes, absolutely. And so normally it's known as the target execution authority, the TEA. And that is the military does a phenomenal job at identifying this well before a conflict when you go forward. And then, so depending on what type of threat that is, it's going to say you have the authorities at your level to accomplish the following things. So if you are an infantry battalion commander, you're a lieutenant

colonel, and you have your soldiers out there on the battlefield, you're only going to have a certain amount of authorities. Your rules of engagement are going to stop at a specific level.

But then as it goes further and further up, what they call the operational command, the operational command. So if you're looking at the Iran scenario, a lot of those have been delegated down to US CENTCOM, US CENTCOM, US Central Command delegates quite a bit to Fifth Fleet so that the US Navy can answer those questions. So if you have a laser onboard a U.S. Navy vessel as part of the George Washington or the Abraham Lincoln Carrier Strike Group, then they will know at their level what they're allowed to do and what they're not. And that's going to be determined based off of that. So it changes. I don't want to say that it's always going to be the following, but the military knows at that point in time. They also know that they have the right to self-defense.

So that that is always priority number one, no matter how you're looking at it. But normally there will be a there will be a ship captain, there will be a carrier strike group, you know, Admiral that will will have that specific authority. And so when he sees it, right? So part of what they end up doing is they put all this stuff into the Aegis system, and they can see all of those. There will be a common operational picture. And based off of that, that Admiral can make that decision. What makes a weapon system exquisite? I hear this term from time to time in these comforts, and I realize I actually don't know the answer to that. So normally it really just comes down to the sophistication and technology that is included in it. So if you're looking at an interceptor, if it is an exquisite interceptor, it means you have the top of the line secret technology that is inside the payload, inside the front of that interceptor, because you're trying to shoot a bullet down with a bullet.

at that point in time, right? The science that's behind that is phenomenal because you have to be able to see exactly where that threat is coming from and whether or not now you're getting into maneuverable systems as well, right? It's not just always a 100% path. I'm flying directly at you, never going to switch path. So these seekers have to be able to then not only do it, they have to identify and track, but then they also have to then do adjust to whatever changes in those paths also come. So that is an exquisite

interceptor type of munition. If you have long range precision munitions that we are firing offensively, Same thing kind of applies it has to do with how fast it goes whether or not the tail fins maneuver these and cause it to to change course as well because now we're trying to also then avoid enemy air defense systems as well so that we can make sure that we hit the target as well. So the payload that's on it how fast how big of an explosion are you trying to achieve.

all of those things playing into that exquisite munitions perspective. Got it. The other side is the gravity bombs. Gravity bombs are just standard GPS guided munitions where as long as you can get over the target and you drop that, it's very, very simple maneuvering and all that. It's a lot less because you're not worried so much about the energetics behind it. Yeah. And you're really just letting it hit the targets. Aren't all bombs gravity in the end? I mean, they all have to like fall back to Earth eventually. Okay, anyway. To what extent do military commanders take into account economic costs when they're making these operational decisions? Like, would someone hesitate to press the button on launching a bunch of ballistic missiles because they're thinking about how much it might cost the US government? And I'm thinking back to, I'm sure we've all seen Top Gun when Maverick...

crashes the airplane or whatever and his commander is like chewing him out and saying that's not your plane it's not your money it's the taxpayers plane and the taxpayers money does that factor into actual operational decisions at all so i think the best way you described it perfectly and i think that that actually shows how great the military system is is because if you are a tactical commander and you have If you're in that situation, all that, you are going to use whatever is necessary to protect your force. It doesn't matter. You have the operational command, and you have those abilities. As you get farther up the food chain, as you become those flag officers, those general officers, the admirals and the generals, when you start getting to those operational higher level of command decisions, they're the ones who are tied more into the national. So you have tactical operational strategic and national authorities.

as you get closer into those strategic and national authorities, that's when those leaders are the ones who start looking at that and saying, I need

systems where I'm not spending 4.1 million to shoot down a 30,000 drone. But that is not something that your standard lieutenant or captain or major is dealing with at their level. Right. I would imagine, though, that they're not thinking about, oh, what is this going to cost the taxpayer? But they are still fundamentally having to think about economizing. They have a certain amount of capacity with them at any given moment, and they have to think when are they going to use their more scarce weapons that they probably does cost more, but scarce weapons. So we have this problem.

Everyone's talked about it. The war in Iran, this rapid depletion of very expensive stocks. Where would we have to get to with some of the weapon systems you're talking about such that or where are we on the path such that these technologies could meaningfully change some of the economics so that we could accomplish some of the same goals that we want while on the very abstract level, saving money, but just generally speaking, solve this problem of ours of munitions, depletement. Yeah, no, I think we're in the middle of it right now because when you look at how the military is structured and how the Department of War, Department of Defense is right now pushing some of these additional, what they call the multi-year framework agreements for these munitions platforms, they're doing that because they're trying to, they're recognizing right now that it has been I don't want to call it a failure of economies of force, but the U.S. military has focused so long on looking at the peer-type fight. So this has been a lesson learned for the military to say, hey, look, I right now, as you even mentioned, here is my arsenal. This is what I have available to me right now. And I'm getting shot at right now. I'm getting attacked by Fatal 110 ballistic missiles. I'm getting attacked by Shahid 136 drones. And hey, what I have at my...

fingertips right now are pack three interceptors from the Patriot system. That's what I have. And so therefore that's what I'm going to use. The Navy's looking at it and they're like, well, I've got SM threes. I've got SM sixes. And so that's what I am using. And so what you're seeing right now is, hey, we don't want to do that. That's been the demand, the demand signal to industry right now is that low cost containerized munitions, those low cost interceptors, the high powered microwave, high energy laser, the urgency and demand for some of these systems is because exactly what

you're saying is I want at my level. If I was sitting right now and I was still Colonel Sanders KFC, you know, sitting at, you know, and I'm working with my force. I want to have all of this solution so that I have more options at my fingertips. I want to have more arrows in my quiver so that I can make the decision between electronic warfare, high-powered microwave, high-energy laser, guns, cheap interceptors, as well as exquisite missiles. I want to have all of those things at my fingertips, and that's what we're at right now.

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How easy or difficult is it to actually produce these things at scale? Because a couple of familiar names kept coming up when I was researching this. Rare Earths, including Gallium. And this one, this one is unfamiliar to me. Joe, do you know what Itterbium is? Itterbium. Itterbium? Why terbium? I think it's Itterbium. No, keep going. All right, well, both of them. as is tradition are mostly produced by China. So it seems like if you want to ramp up production of these things, even if you agree that these are effective weapons and they're going to do a lot for defensive capabilities, scaling up production could still be a challenge. No, it absolutely is. When you look at critical minerals across, there's terbium, like you mentioned. There's dysprosium. Dysprosium is another one that a lot of people use. Military uses a lot of new denium boron iron magnets. And the fact that I know that a thumb ahead just shows how many times we've had it, we've had to go over this. But yeah, so critical minerals becomes very, very key. And so when you start looking at some of the deals that are being made by the Department of War, right now is a great example of that, right? The deal with MP materials to actually, you know, gain a piece of ground specifically inside the US for domestic onshoreing of these capabilities. But you're

absolutely right. There are quite a few of these critical minerals that China has.

a predominance in the world. They have sometimes 85, even 90% of the supply chain. So how important it is that we're using the ones that we have correctly when China last year added 12 critical minerals that they were not going to share with the United States. These are some of the key ones, like you said. And so the nice thing is that a lot of the defense primes have recognized this as a risk for quite a while. And so they've already been working towards some of these. But the bigger problem now is these are not just critical minerals that are used for defense platforms. These are a lot of dual use capabilities with a lot of civilian and commercial needs as well. And so if you're looking at a brushless motor for a drone, that could be used for a civilian drone or it could be used for a military reaper drone or a global hawk. And so when you start looking at those, it is a lot of decisions that have to be made by industry.

By the way, I have to let listeners know who've been listening for a while. I found my tungsten cube. Thanks to Tracy. I had been missing it. And she suggested a drawer that we store stuff in that it might be in and it was there. So thank you. Trento, you can buy a Turbium cube. Oh, no. No, it's 40 for a little 10 millimeter cube. Does it look nice? It's just so great. Yeah, it's just a. It's just a silvery cube. I'm curious, like when like maybe a big picture when like a sort of new type of like weapons regime sort of maybe disruptive weapons tech emerges in your seat as an analyst, how much of this is coming from the primes and how much of this is coming from defense startups, etc. There's like some defense startup is like, you know, we don't have an exquisite missile system that we want to keep selling. And therefore, we're not going to fall captive to the innovators dilemma, unlike a defense prime that may not be as motivated to push on a technology that could, you know, render their existing workhorse sales to be less market competitive. What do you see in terms of market dynamics here? No, it is a great point, because you're hitting on something that the DOW has actually identified now and has started to push towards as well. The defense primes right now are scaling. We've already talked about the exquisite munitions, right? And so a lot of the multi-year framework agreements that have come to them has been to increase the stockpiles of things that we already have.

So pack three interceptors going from 620 a year to 2000 per year, right? That is what they're pushing.

That's a 3.3x increase on what they're doing. The THAAD system, the Terminal High Altitude Air Defense System. These are like 12 to 15 million per interceptor. They're trying to increase that 4x. And then you've got the Precision Strike Munition, which is a long-range offensive system that we've used in Iran successfully. That's very strategic to like a China fight. That is one that they're trying to increase by up to 4x as well. So you start putting those together, that's your exquisite munitions piece. But that middle ground that I was talking about, right, where you're looking for these low-cost containerized munitions, you're looking for, okay, I don't have, I'm not ready to just do a high-powered microwave, but I don't want to use a 4 million interceptor. There are contracts that are out right now, specifically, five companies, Coaspire, Lightos, Andrewle.

Zone 5 and Castilian are the five contracting companies right now that the DOD has contracted with and say, I need this middle ground solution. I need tens of thousands of these. Not trying to jump like I said from 620 to 2,000 or 96 to 400 for the THAAD system. We're talking about I need thousands, tens of thousands. That's my demand signal. Let's make this happen. And so the market, these disruptors are the ones who are picking up those contracts. So the short term, the primes are winning because they already have the production lines. They are able to put these contracts directly into place and they're pushing for it. But that mid-grade, think about three to five years, the scalability for the Andarals and the Castilians of the world to come in and grab that middle market as well is an area that the primes aren't looking to grab into too much.

and they're allowing some of these tech disruptors to do it, but you're also seeing a lot of joint ventures as well across the board, not only internally to the U.S., but also our partnerships around the world. Lockheed does a great job, so does RTX, partnering with some Europeans, companies as well, Reimatol being one of them, so that they can start to increase capacity overseas as well. Can you talk a little bit more about, I guess, AI and... safety issues when it comes to this kind of weaponry because Joe and I, we've been having a lot of conversations recently about LLMs and some

recent safety incidents, including an open AI model escaping its sandbox during a test and hacking into hugging face, which is like bad enough. But then when you think about AI potentially in a military environment and maybe having access to a bunch of ballistic missiles or now laser beams, that seems kind of terrifying.

No, absolutely. And funny enough, when I was at war college, my group did an entire thing about AI and levels of trust because you really are looking at a lot of the stuff that you're originally seeing partnered into the Pentagon right now is things that are going to help us do our staff jobs better. Let's look at how to be able to optimize processes. How do we do a better job of tagging our capabilities so that we can track them better and so on and so forth. What it hasn't entered into very much is really when you look at identifying friend and foe leading to a command decision for a tactical strike. Those are things that are still maintained. Those are human in the loop AI problems. And so that's really where you're seeing a lot from the military side right now is I want you to help me clean up the noise. You see a lot of that on the imagery intelligence side of the house and saying, okay, I want AI to be able to look through a lot of my platforms. I want to take all the satellite imagery. I want you to help me normalize this and help me see where the change detections may be. Pickaxe Mountain would be a good example of that. I'm using this as a theoretical, but let's say we wanted to focus on everything and anything that's going around Pickaxe Mountain for the nuclear Iran discussion.

So AI could potentially do that from an imagery perspective to say, all right, I've watched the videos for 12 to 16 hours and based off of it, this is what we have and you can flag an alert. but that alert for something that's going to happen. Maybe it's a dump truck driving away, and that's what you told it to look for. Great, but there's still going to be a human that's looking at that then. When that alert comes on, there's an analyst that's sitting there. There's a commander that would make a decision behind that that says, this is what I want to do about that truck that is driving away, where we are not at right now and where we are quite a ways away based off of trust models and all that. I think exactly where we should be.

is we are not at the point where AI is going to go I see that truck I've been given in order to destroy it and I am going to take that target execution authority that I mentioned earlier and just drop the bomb. So when I'm looking at one of your most recent research notes on the Bloomberg terminal and you talk about the directed energy market growing two to three times from where it's at currently which is six to eight billion dollars and that's over the next decade. I just want to be clear. Is that a forecast for like. actual procurement or is that including R&D on the whole? Because it seems like so far most of the money that's been spent has been on development.

Absolutely. I mean, we have spent over 15 billion on the R&D up until this point, specifically to get to where we're at right now. And it is a good news story because it is transitioning to procurement. You are starting to see that as different portions. The Army, like I mentioned earlier about the Striker vehicles, the 50 kilowatt laser platforms and all that. You're starting to see that start showing up inside of the multi-domain task forces. You're seeing the Navy continue to push for additional high-energy laser solutions. Contracts with Epirus, one of the companies that does some high-powered microwave stuff, is tied directly into both the Army and to the Marine Corps because of their capabilities to protect against drone swarms. And they've even developed something called the Leonidas H2O system, which helps disable boat motors. So when you start looking at some of the tactical problems that you run into in the Black Sea and the Red Sea and all that for just even remote controlled.

fast boats trying to go after blue water Navy assets that cost hundreds of millions if not billions of dollars. Having these types of capabilities are exactly what you're looking for. So right now is some of the transition from the R&D side into the procurement market. But you are correct. There still is a lot more R&D because if you look at the research and engineering department from the Pentagon. They're looking for a one megawatt laser by 2030. The science continues to go, right? It's really questions to whether or not the physics and the science behind can continue to support and bring down those size, weight, and power requirements necessary to be able to produce it. I have to say, and I've said this on many episodes, I do not have great intuitions.

for any time we're talking about units of electricity whether it's megawatt or megawatt hours the context of a grid can you give me an a when you say they're looking for a one megawatt laser What are what is that? How should I conceptualize this? What's a metaphor that I could use to understand what that would mean? Yeah, so right now the 50 kilowatt laser is capable of if a drone is flying at you should be able to knock that drone down At one time right from a kilowatt perspective when you start looking at one megawatt You're trying to create a beam that is so strong that you can shoot down something that is flying at you at you know a thousand miles an hour potentially up to four to five thousand miles an hour as is in Mach five hypersonic weapon wood. And the gap between getting from 50 kilowatt to one megawatt is that on the sort of energy side that's on the side of the laser? Is it a technology that exploits that much energy? Like what needs to be solved to make that jump?

Yeah, I would say it's almost an iterative thing right now. DARPA is looking for, you know, higher level of solutions. They're looking for almost a revolution of military affairs because they want to jump iterations or increments in terms of what the development is. But I think it's a little bit of all of those, right? Part of it is on the physics side of the house to be able to generate one megawatt and keep the beam where you want it to be. But then also, it's also deployability discussion. Having a one megawatt laser that is static because you need to plug it into shore power is limited in a certain aspect. It is great for self-defense and force protection. So think about it in terms of putting it on a military base in Kuwait. That would be a great opportunity to be able to do it because that is a shore power solution with a fairly unlimited magazine that's attached to it. But when you start looking at a potential peer adversary threat where you want to be able to move your forces forward and you have to you know have to contend with these types of missiles being shot at you you're not it's very difficult to carry around a generator that can produce a one megawatt capability that you then use to create these impulse beams. Just going back to Israel's iron beam or laser dome for a second is that basically a test case for a lot of this technology and how it might be useful in a real world environment. And then if so, what are you actually looking out for in terms of its operations to prove that it's economically scalable or actually useful?

Yeah, iron beam is kind of leading the way, right? It's about, I think, if I'm not mistaken, it's about 150 kilowatts compared to our 50 kilowatts. So when you start looking at how they deploy it, how they use it, some lessons learned that come from them, I think it is an operational test bed as they are being shot at in the use operationally to say, okay, I successfully hit a Shaheed 136 drone from this distance. How far away did I shoot it? How long did it take for me to actually get that thing to stop coming at me? These are all things that a lot of partners with Israel are looking at specifically so that they can identify, all right, I need to make the following changes in my research and development side of the house. One of the things on lasers, we didn't even talk about a little bit, is some of the shortcomings of it. Because one of the key things that's there is atmospheric does affect.

how that system actually works. A missile doesn't care if you're shooting through fog or through rain, but it does have a change to the components, if you will, of how a laser beam is actually going to go out there. All right, Wayne Sanders, we're going to get kicked out of the studio soon, so we're going to have to leave it there. But thank you so much for coming on OddLots. Thank you for having me. It was great. So Joe, very interesting conversation. One thing that stands out to me is just... That pure adversary miscalculation. Yeah. Like it is crazy to think how much the US optimized its military for similar militaries, right? We're going to stock up on like the best technology, the most exquisite missiles and all of that. And then they find themselves fighting an adversary that is using cheap drones and suddenly the whole balance is thrown off.

This came up, I mean, just this general idea of military planning for types of warfare. It seems very difficult. This came up in the recent episode we did with David Fickling about tungsten and the military concluded, well, future wars, it doesn't look like tungsten is going to be a bigger thing. And then suddenly you get wars like the war in Ukraine, which more resemble wars in which you just needed a big heavy metal, etc. I have an important question. Yeah. Will you donate your tungsten cube to the military? We're deleting this. That's a critical millet mineral. If they drive truly comes and they're like everyone bring your tungsten or something like that, you know. I don't wanna, I'll be like, sure. Take the cube. Imagining all these mostly

bros walking down the streets of Manhattan. Handing their tungsten cube. But no, I found that to be really interesting just in general because laser weapons does seem like one of these things that...

Again, we've been seeing in movies forever. You just shoot the laser beam, but thinking about the actual constraints he mentioned, you know, these shore power systems, etc. Think about the actual constraints of firing this like incredibly amount of directed energy in certain environments, then thinking about weather constraints, etc. It gets super interesting. Also, I'm just very jealous of the guys that can rattle off all of these. You're very good at it as well. At what? rattling off the names of weapons systems and so far. No, well, I'm cheating because I have them all written down on my screen because otherwise I forgot. No, that's like a very impressive. Trust me. I don't think about like coyote and vampire systems on a daily basis. Well, even though that those were the cops that we should be asking about, I'm very impressed. Well, the one thing the one thing I'm thinking is the cost effectiveness of all of this. So.

One way you often see it framed is the laser beams, the microwaves are sort of like the renewable energy of the weapons world in that once you create them, you can continue firing off. It's that deep munition idea versus the sort of fossil fuel ballistic missiles that are finite. Right. But as we all know, sometimes installing renewable energy systems is extremely expensive to the point where a lot of countries and cities decide not to do it. think we still have to see on that. That's totally the scarcity move somewhere else. It may be optimal in some conditions, it may not. But there is still this fundamental scarcity trade off of okay, like you might have lots of ability to not deplete, but then it much be much might be much harder to install the system. Absolutely. It does not seem like there is a silver bowl a magic bullet out there for these questions. That's a good pun. All right, shall we leave it there? Let's leave it there.

This has been another episode of the OddLots podcast. I'm Tracy Allaway. You can follow me at Tracy Allaway. And I'm Joe Wiesenthal. You can follow me at the stalwart. Follow our producers, Carmen Rodriguez at Kerman Arman, Dashel Bennett at Dashbot, Kale Brooks at Kale Brooks, and Kevin Lozano at Kevin Lloyd Lozano. And for more OddLots content,

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